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Composite Identifiers for Wearable Articles

Abstract

An article includes: a first portion having a first graphic; a second portion having a second graphic, the second portion being translucent; a body extending between the first portion and the second portion, the body being deformable to affix the second portion at a position overlaid with the first portion, wherein the second graphic is configured to overlay with the first graphic to define a composite graphic corresponding to the position.

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Background/Summary

BACKGROUND

[0001] Controlling access to physical facilities, services such as software accessed via computing devices, or the like, may involve identifying each party seeking to access the facility and/or service

uniquely from other parties. In some environments, the provision of unique identifiers to potentially large numbers of such parties may be costly and/or time-consuming.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

[0002] The accompanying figures, where like reference numerals refer to identical or functionally similar elements throughout the separate views, together with the detailed description below, are incorporated in and form part of the specification, and serve to further illustrate embodiments of concepts that include the claimed invention, and explain various principles and advantages of those embodiments.

[0003] FIG. 1 is a diagram of a system for managing access to an access-controlled resource.

[0004] FIG. 2A is a diagram of an identification article in the system of FIG. 1.

[0005] FIG. 2B is a cross-section of the article of FIG. 2A.

[0006] FIG. 2C is an assembled view of the article of FIG. 2A.

[0007] FIG. 3 is a diagram of detailed views of the first and second graphics of the article of FIG. 2A.

[0008] FIGS. 4A, 4B, and 4C are diagrams of distinct composite graphics generated by the article of FIG. 2A.

[0009] FIG. 5 is a diagram illustrating the generation of a unique identifier from the composite graphics shown in FIGS. 4A-4C.

[0010] FIG. 6 is a flowchart of an authentication method using the identifiers generated as illustrated in FIG. 5.

[0011] Skilled artisans will appreciate that elements in the figures are illustrated for simplicity and clarity and have not necessarily been drawn to scale. For example, the dimensions of some of the elements in the figures may be exaggerated relative to other elements to help to improve understanding of embodiments of the present invention.

[0012] The apparatus and method components have been represented where appropriate by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present invention so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein.

DETAILED DESCRIPTION

[0013] Examples disclosed herein are directed to an article, comprising: a first portion having a first graphic; a second portion having a second graphic, the second portion being translucent; a body extending between the first portion and the second portion, the body being deformable to affix the second portion at a position overlaid with the first portion, wherein the second graphic is configured to overlay with the first graphic to define a composite graphic corresponding to the position.

[0014] Additional examples disclosed herein are directed to a method, comprising: capturing an image of an article having a first graphic overlaid with a second graphic to define a composite graphic, the composite graphic including a boundary of the second graphic, marks of the second graphic, and a portion of a plurality of marks of the first graphic; detecting, from the image, the boundary of the second graphic; generating an identifier based on the composite graphic; and executing an authentication action based on the identifier.

[0015] FIG. 1 shows a system **100** for managing access to an access-controlled resource **104**. The resource **104** can be an enterprise software application hosted on one or more computing devices, e.g., connected with a network **108**. In other examples, the resource **104** can be a facility (e.g., a building, complex of buildings, or the like), such as a healthcare facility, an entertainment venue, or

the like. A person **112**, e.g., a patient or staff member at healthcare facility, an employee at a warehouse, an attendee at an entertainment venue or the like, may be permitted to access the resource **104** upon completing an authentication process, e.g., implemented by an authentication server **116** that can be implemented as a computing device connected with the network **108**.

[0016] The authentication server **116** can, for example, store a plurality of account records each corresponding to distinct individuals. Each account record can include various information corresponding to a given person, such as an account name and/or other identifier, a secret such as a password and/or encryption key, and the like. The authentication process serves to determine whether a person purporting to be a holder of an account maintained at the server **116** is truly the account holder.

[0017] Some authentication processes involve the provision of an account identifier and password by the person **112** to the server **116**, e.g., via a client computing device **120** operated by the person **112**. Certain authentication processes involve the provision of additional information, such as biometric authentication, a personal identification number (PIN), a message from a further computing device also operated by the person **112**, or the like.

[0018] Other authentication processes may involve assigning a unique identifier to the person **112**, and subsequently verifying that the person **112** is in possession of that unique identifier. The unique identifier can be provided to the person **112**, for example, in the form of a wristband or other wearable article that carries the unique identifier, e.g., in the form of a barcode or other machine-readable form. A unique identifier can also be used to supplement other authentication processes, for example to facilitate re-authentication by the person **112** and thereby regain access to the resource **104** without repeating the provision of an account identifier, password, biometric data, or the like. That is, in an initial authentication process the person **112** may supply authentication data such as an account identifier and a password, as well as the unique identifier, and subsequently (e.g., for a time period determined at the server **116**) the person **112** may regain access to the resource **104** by presenting the unique identifier without account identifiers, passwords, and the like.

[0019] Although the use of a unique identifier may reduce the time spent by the person **112** to authenticate with the server **116** and gain access to the resource **104**, providing unique identifiers within the system **100** may be costly and/or time-consuming. For example, in some systems unique identifiers may be predetermined and stored at the server **116**, as well as deployed to users of the system **100** (such as the person **112**) via distinct physical tokens, such as wristbands with barcodes or the like each encoding distinct identifiers. The generation and storage of unique identifiers, as well as the manufacturing, deployment, and securing of physical tokens with distinct identifiers, can be costly.

[0020] To mitigate the impact of deploying individually-assigned unique identifiers for users of the system **100** such as the person **112**, the system **100** includes a plurality of articles **124** that need not encode any specific identifier in an initial state. In some examples, the articles **124** can be substantially identical to one another in an initial state, e.g., prior to use by the person **112**. The articles **124** are configured to dynamically generate unique identifiers when used, as discussed below. The articles **124** can be manufactured in bulk, without any predetermination of unique identifiers, and supplied to a facility or other location where the person **112** and other users of the system **100** seek access to the resource **104**. Upon use, each article **124** can dynamically generate a unique identifier that distinguishes that article **124** from the other articles **124**. The dynamically generated unique identifier can be used for a variety of purposes, including enrollment at the server **116** in association with the person **112** (e.g., via the client computing device **120**) and subsequent use to grant access to the resource **104** for the person **112**.

[0021] Certain components of the client device **120** are illustrated in FIG. 1. The client device **120** includes a processor **128** (e.g., a CPU, GPU, and/or other suitable control circuitry, microcontroller, or the like), and a non-transitory computer-readable medium such as a memory **132**. The client

device **120** also includes a communications interface **136** configured to communicate with other computing devices via the network **108**, such as the authentication server **116** and, in embodiments where the resource **104** is a hosted application, the resource **104**. In an embodiment, the resource **104** is an access-controlled application server configured to accept data captured by the client device **120**.

[0022] The client device **120** may further include one or more input devices **140** and one or more output devices, such as a display and touch screen assembly. The client device **120** can include other input/output devices, such as speakers, microphones, keypads, barcode scanners, radio frequency identification (RFID) readers, and the like. The client device **120** also includes a camera **144**. As discussed below, the camera **144** can be controlled to capture an image of an article **124**, and determine a unique identifier for the article **124**. The unique identifier can be enrolled with the server **116** for later use in authenticating or re-authenticating the person **112**, for example. The memory **132** can store computer-readable instructions, e.g., including an authentication application **148**, executable by the processor **128** to implement functions related to the capture and processing of images of the articles **124** to derive unique identifiers. In an embodiment, the camera **144** may also function as an input/output device **140**.

[0023] Turning to FIGS. 2A, 2B, and 2C, an example article **124** is shown. The article **124** in the illustrated example is a wristband, although the articles **124** can take other forms in addition to or instead of wristbands. Other example form factors for the articles **124** include arm bands, belts, lanyards, plant tags, and the like. It is also contemplated that while the articles **124** are discussed herein in the context of providing identifiers for users of the system **100** such as the person **112**, the articles **124** can also be employed to identify objects such as parcels, animals, machinery components, trees, plants, or the like.

[0024] The article **124**, as shown in FIG. 2A, includes a first portion **200**, a second portion **204**, and a body **208** extending between the first and second portions. In this example, the first portion **200**, the second portion **204**, and the body **208** are formed from a single piece of material, e.g., a suitable wristband substrate such as a flexible polymer. In other examples, any one of, or any combination of, the portions **200** and **204** and the body **208**, can be formed from distinct materials joined together by adhesives or other suitable fasteners. In some examples, the body **208** can include additional components, such as a bracket or other support for an identification tag, a coating of material configured for printing of indicia thereon, or the like. The body **208** can, in other words, support a data carrier **209** such as an RFID tag, a barcode, a label including text, images, or the like. The data carrier **209** can be disposed on an outer surface of the body **208**, as illustrated, or can be integrated within the substrate forming the body **208**. The data carrier **209** can store, encode, and/or display a unique identifier, for example. In an embodiment when the article is used, the body **208** may be configured to support an active radio frequency identification tag, such as an ultrawideband location tag or a Bluetooth low energy (BLE) tag.

[0025] The first portion **200**, in the illustrated example, is disposed at a first end **210** of the article **124**, and the second portion **204** is disposed at an opposite end **211** of the article **124**. Further, the first portion **200** includes a first graphic **212**, and the second portion **204** includes a second graphic **216**. As seen in the cross-sectional view of FIG. 2B, the graphics **212** and **216** can be formed on the substrate material defining the body **208** and the portions **200** and **204** as a coating of ink or other pigment (the thickness of which is exaggerated in FIG. 2B for illustration). In other examples, the graphics **212** and **216** can be embedded within the substrate material.

[0026] In the illustrated example the article **124** further includes a fastener, such as an adhesive **220**, shown in FIG. 2B on an opposite side of the portion **204** from the graphic **216**. The adhesive **220** permits the second portion **204** to be affixed to the first portion **200**. In particular, in the illustrated configuration, the body **208** can be deformed, e.g., to encircle a wrist of the person **112**, and the adhesive **220** can be placed over the first portion **200** to form the article **124** into a loop.

[0027] The adhesive **220**, in other words, can be placed onto the graphic **212**, such that the second

portion, including the second graphic **216**, is overlaid on the first graphic **212**, as shown in FIG. 2C. While the first portion **200** and the first graphic **212** are covered by the second portion **204**, as indicated by dashed lines in FIG. 2C, at least the portion **204** of the article **124** is translucent and the first graphic **212** remains visible through the second portion **204**. In some examples, the substrate (e.g., the body **208** and the portions **200** and **204**) is translucent or transparent. In other examples, the portion **204** is translucent or transparent, while the body **208** and the portion **200** can be opaque. That is, the portion **204** and the adhesive **220** permit the passage of sufficient light (e.g., visible light) therethrough for an image (e.g., captured by the camera **144**) depicting the article **124** in an assembled configuration as shown in FIG. 2C to show the second graphic **216**, and at least a portion of the first graphic **212** underneath the second graphic **216**. The second graphic **216** and the overlaid portion of the first graphic **212** form a composite graphic that can be employed, e.g., by the client device **120**, to generate an identifier for the article **124**. FIG. 2C also illustrates the article **124** in an overhead view, showing the graphics **212** and **216** and the adhesive **220** in an assembled configuration. The thickness of the graphics **212** and **216**, and the adhesive **220**, are exaggerated for illustrative purposes.

[0028] In some examples, the adhesive **220** need not be transparent, but can instead be disposed at either end of the second portion **204**, rather than directly underneath the second graphic **216**. In further examples, the second graphic **216** can be defined by pigmented adhesive, such that a single coating in the position of the adhesive **220** as shown in FIG. 2B can provide both the second graphic **216** and the adhesive **220**.

[0029] Turning to FIG. 3, the first and second graphics **212** and **216** are shown in greater detail. The second graphic **216** can have a surface area that is smaller than a surface area of the first graphic **212**. The second graphic **216** further includes a boundary **300**, such as a solid line in this example. In other examples, the boundary **300** can be defined as a set of concentric arcs, as a dashed line, or the like. In an example, the boundary **300** is a set of three or more connected line segments of different lengths, where the ratio of the length of two segments is an integer. The second graphic further includes a plurality of marks **304**, such as line segments at various angles of orientation. The second graphic **216** can include a wide variety of other forms of marks, such as dots, polygons, zebra stripes, and the like. The second graphic **216** further includes a background **308** (e.g., the portion of the second graphic **216** within the boundary **300** and not occupied by the marks **304**) that is translucent or transparent. The boundary **300** and the marks **304** can be opaque.

[0030] The first graphic **212** can include a boundary **312**, such as a solid line, a dashed line, a set of concentric arcs, or the like. In some examples, the boundary **312** can be omitted. Within the boundary **312**, the first graphic **212** includes a plurality of marks **316**, which in the illustrated example are lines extending from one side of the boundary **312** to an opposite side of the boundary **312**, substantially perpendicular to the edges of the article **124**. The marks **316** can have a variety of other angles than those shown, and need not be at the same angles. The marks can also include dots, polygons, zebra stripes, or the like. In some examples, such as that shown in FIG. 3, the marks **304** are oriented within a threshold angle (e.g., thirty degrees, though other thresholds are also contemplated) of the edges of the article **124**, while the marks **316** are oriented within a threshold angle (e.g., one hundred twenty degrees, though other thresholds are also contemplated) of the ends **210** and **211** of the article **124**. The marks **304** are therefore more likely to be perpendicular to the marks **316** than parallel when the article **124** is assembled for use. When the second graphic **216** is overlaid on the first graphic **212**, intersections between marks **304** and marks **316** are likely to result. Such intersections may facilitate the generation of a unique identifier for the article **124**.

[0031] The second graphic **216** can have a predefined color (e.g., red, in some examples, although any of a wide variety of colors can be employed), and can also include various other predetermined attributes, such as predefined dimensions for the sides of the boundary **300**. The first graphic **212** can also have a predefined color, e.g., different from the color of the second graphic **216** (e.g.,

cyan, in some examples). In other examples, each graphic **212** and **216** can include more than one color. The provision of distinct colors for the graphics **212** and **216**, or in some examples for at least the boundary **300**, may facilitate the detection of the boundary **300** within an image of the article **124**.

[0032] When the body **208** of the article **124** is deformed to affix the second portion **204** to the first portion **200**, the specific location of the second graphic **216** relative to the larger first graphic **212** is variable, e.g., depending on a size of the resulting loop formed by the article **124**, as well as the angle at which the ends of the article **124** are aligned before affixing the second portion **204** to the first portion **200**. For example, a person with a smaller wrist may affix the second portion **204** at a different position on the first portion **200** than a person with a larger wrist.

[0033] Turning to FIGS. **4A**, **4B**, and **4C**, affixing the second portion **204** at a given position relative to the first portion **200** generates one of a plurality of composite graphics **320**. Three example composite graphics **320-1**, **320-2**, and **320-3** are shown in FIGS. **4A**, **4B**, and **4C**, and are collectively referred to as composite graphics **320**, or generically as a composite graphic **320**. The composite graphics **320** are defined by the boundary **300**, the marks **304**, and a subset of the marks **316** that appear within the boundary **300**.

[0034] As shown in FIG. **4A**, affixing the second portion **204** at a first position relative to the first portion **200** generates a composite graphic **320-1**, containing certain ones of the marks **316**. As shown in FIG. **4B**, affixing the second portion **204** at another position relative to the first portion **200** generates a different composite graphic **320-2**, containing a different subset of the marks **316** than the composite graphic **320-1**. FIG. **4C** shows a further example, in which the second portion **204** is affixed to the first portion **200** at an angle (that is, the first and second portions **200** and **204** are not substantially parallel, as in FIGS. **4A** and **4B**). The position shown in FIG. **4C** yields a further composite graphic **320-3**. The composite graphic **320-3** contains the same subset of the marks **316** as the composite graphic **320-2**, but that subset of marks **316** is oriented differently in the composite graphic **320-3** than in the composite graphic **320-2**.

[0035] It will be understood that the second portion **204** can be affixed to the first portion **200** in a wide variety of other positions, each yielding a distinct composite graphic **320**. A single article **124** can, in other words, generate any of a wide variety of composite graphics **320**, and thus a supply of initially identical articles **124** can generate a plurality of distinct identifiers.

[0036] The use of a fastener such as the adhesive **220** may prevent the relative positions of the first and second graphics **212** and **216** from changing after the second portion **204** is affixed to the first portion **200**. The article **124**, in other words, can generate one of a variety of composite graphics, and that composite graphic is then unlikely to change. The composite graphic can be used to generate a substantially unique identifier for the article **124**. Therefore, a plurality of articles **124** that initially (e.g., before use) are substantially identical can generate, when worn by users such as the person **112**, a plurality of different identifiers that can then be used for authentication or other identification functions. In embodiments employing adhesives as fasteners, returning the article **124** to an initial state, in which the first and second portions **200** and **204** are no longer overlaid, may be impossible, or may damage the article **124**. The article **124**, in other words, may be tamper-evident. Further, removal of the article **124** by stretching and/or disengagement of the adhesive **220** destroys the composite graphic **320**, and recreating a substantially identical composite graphic **320** may be difficult or impossible.

[0037] Turning to FIG. **5**, the generation of a unique identifier from an article **124** is discussed in greater detail. The generation of the unique identifier can be performed, for example, by the client device **120** via execution of the application **148**. The client device **120** can be controlled, e.g., by the person **112** or another operator, to place at least a portion of the article **124** bearing the composite graphic **320** within a field of view of the camera **144**. Via the camera **144**, the client device **120** can capture an image **500** of the article **124**, depicting the composite graphic **320**. The client device **120** can be configured to detect the boundary **300** in the image **500**, e.g., based on

predefined dimensions of the sides of the boundary, the color of the boundary, the relative size of portions of the boundary **300**, and the like. In an embodiment, a trained computer vision model may detect the boundary **300**. The client device **120** can be configured to extract the composite graphic **320** from the image **500** based on the detected boundary **300** (e.g., by discarding any portion of the image **500** that is outside the boundary **300**).

[0038] The client device **120** can then be configured to generate a unique identifier **504**, e.g., in the form of a numerical representation of a set of features of the composite graphic **320**. Various mechanisms can be employed to generate the identifier **504**. For example, the client device **120** can detect intersections between the marks **304** and **316**. In other examples, the client device **120** can detect corners or other keypoints within the composite graphic **320**, e.g., via the Scale-invariant Feature Transform (SIFT) keypoint detector algorithm or any other suitable keypoint detector. The keypoints selected by the client device **120** may be substantially invariant to scale, orientation, and lighting artifacts, to permit later detection in distinct images of the article **124**.

[0039] The client device **120** can then determine, for each keypoint, a numerical representation such as a binary descriptor vector. Various descriptor generation mechanisms can be employed for this purpose, including for example the Oriented FAST and Rotated BRIEF (ORB) descriptor. The ORB descriptor operates on intensity values and can therefore include converting the composite graphic **320** to grayscale, but the client device **120** can also implement other descriptor generation mechanisms that operate on color images. In an embodiment where a first graphic **212** is red and a second graphic **216** is cyan, the descriptor mechanism may operate on a first monochrome (cyan) color plane to describe attributes of the second graphic and a second color plane to describe related attributes of the composite image. Generating a binary descriptor can include, for a given keypoint (e.g., a specific pixel or a patch of pixels), sampling a plurality of pixel pairs of the composite graphic **320** within or around the keypoint (e.g., sampling from a 15×15 pixel grid surrounding the center of the keypoint). For each pair, a binary value is selected based on whether the first pixel in the pair has a higher or lower intensity than the second pixel in the pair. The descriptor for the keypoint is therefore a vector of binary values. The identifier **504** can therefore be a combination of each keypoint descriptor vector, e.g., in the form of a one-dimensional or multi-dimensional array of numerical values. For example, the client device **120** may select fifty keypoints, generate a 256-bit descriptor vector for each keypoint, and concatenate the descriptor vectors to generate the identifier **504**. A wide variety of other forms of identifier **504** can also be employed, however.

[0040] In some examples, the client device **120** can perform additional processing on the composite graphic **320** to generate the identifier **504**. For example, the client device **120** can adjust color and/or intensity values of the composite graphic **320** to calibrate the boundary **300**, marks **304**, and marks **316** to predetermined colors (e.g., red and cyan, as mentioned earlier) before generating the identifier **504**. The background color between the marks **304** and **316**, which may represent clothing or skin of the person **112**, may therefore also contribute to the identifier **504** in a way that is reproducible under different imaging conditions. In an example where the article **124** is assembled around an appendage (e.g., a wrist or ankle) of a human person **112**, skin that is visible within the background area **308** of the composite mark will be a human skin tint within a cyan-red colorspace; in this example a red first mark with a cyan second mark may define a composite identifier, while the red and cyan layers may be used to calibrate a colorspace such that the tint may be quantified within the cyan-red colorspace. That tint may then be used as a biometric identifier for the wearer **110** of the article **124**. For example, in addition to identifying the boundary **300**, the marks **304**, and the marks **316**, the device **120** can identify the background **308**, determine a color value of the background **308** (e.g., after calibration as noted above), and store the determined color value as a component of the identifier derived from the composite graphic **320** or as an independent identifier. In other examples, the device **120** can identify the boundary **300**, marks **304**, and marks **316**, and discard (e.g., overwrite with white, black, or the like) any other pixels prior to generating the identifier **504**.

[0041] By applying the same keypoint detection and descriptor generation mechanisms, the client device **120**, and/or other computing devices, can capture subsequent images of the article **124** and generate identifiers therefrom that either match the identifier **504**, or are sufficiently close to the identifier **504** to indicate a high likelihood that the same article **124** is represented in the subsequent images as in the image **500**. In other words, an identifier that differs from the identifier **504** by more than a threshold is likely to correspond to a distinct article **124**.

[0042] Turning to FIG. **6**, a method **600** is illustrated, e.g., performed by the client device **120** to employ articles **124** for authenticating the person **112** and/or other users of the system **100**. At block **605**, the device **120** can capture an image, such as the image **500** discussed in connection with FIG. **5**. At block **610**, the device **120** can detect the boundary **300** of the second graphic **216**, and extract the composite graphic **320** from the image **500**. At block **615**, the device **120** can generate the identifier **504** based on the composite graphic **320**, as also discussed in connection with FIG. **5**. The identifier generated at block **615** can include, in some examples, generating a color value based on the background **308** and corresponding to the skin of the wearer of the article **124**.

[0043] Once the device **120** has generated the identifier **504**, the device **120** can perform one or both of an enrollment function and an authentication function (which may be referred to collectively as authentication actions), e.g., depending on a mode of operation selected by the person **112** or another operator of the device **120**. At block **620**, the device **120** can select an operating mode, or receive input data selecting an operating mode.

[0044] For example, in the enrollment mode, the identifier **504** may not have been previously associated with the person **112**, and therefore at block **625** the device **120** can collect and transmit and/or store authentication data in association with the identifier **504**. For example, enrollment can include capturing biometric data of the person **112**, prompting the person **112** for an account name and password, or the like (collectively, referred to as authentication data), and transmitting the authentication data and the identifier **504** to the server **116**.

[0045] In the authentication mode, rather than enrolling a new identifier in association with the person **112**, the device **120** can be configured to determine whether the identifier from block **615** matches a previously enrolled identifier. At block **630**, the device **120** can retrieve a reference identifier, e.g., via a search of a repository of previously enrolled identifiers stored at the server **116** or the like. When a matching reference identifier is located that is sufficiently similar to the identifier from block **615**, the determination at block **635** is affirmative, and at block **650** the client device **120** can grant access to the resource **104**. When no matching reference identifier is located that matches the identifier from block **615**, the device **120** can instead deny access at block **645**. The determination at block **635** can include, for example, computing a cosine similarity between numerical vectors representing the identifier from block **615** and the reference identifier. Other suitable matching mechanisms will also occur to those skilled in the art. The device **120** (or the server **116**) can apply a threshold, e.g., such that the reference identifier need not match the identifier from block **615** exactly to result in an affirmative determination at block **635**.

[0046] In the foregoing specification, specific embodiments have been described. However, one of ordinary skill in the art appreciates that various modifications and changes can be made without departing from the scope of the invention as set forth in the claims below. Accordingly, the specification and figures are to be regarded in an illustrative rather than a restrictive sense, and all such modifications are intended to be included within the scope of present teachings.

[0047] The benefits, advantages, solutions to problems, and any element(s) that may cause any benefit, advantage, or solution to occur or become more pronounced are not to be construed as a critical, required, or essential features or elements of any or all the claims. The invention is defined solely by the appended claims including any amendments made during the pendency of this application and all equivalents of those claims as issued.

[0048] Moreover in this document, relational terms such as first and second, top and bottom, and

the like may be used solely to distinguish one entity or action from another entity or action without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” “has,” “having,” “includes,” “including,” “contains,” “containing” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises, has, includes, contains a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a”, “has . . . a”, “includes . . . a”, “contains . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises, has, includes, contains the element. The terms “a” and “an” are defined as one or more unless explicitly stated otherwise herein. The terms “substantially”, “essentially”, “approximately”, “about” or any other version thereof, are defined as being close to as understood by one of ordinary skill in the art, and in one non-limiting embodiment the term is defined to be within 10%, in another embodiment within 5%, in another embodiment within 1% and in another embodiment within 0.5%. The term “coupled” as used herein is defined as connected, although not necessarily directly and not necessarily mechanically. A device or structure that is “configured” in a certain way is configured in at least that way, but may also be configured in ways that are not listed.

[0049] Certain expressions may be employed herein to list combinations of elements. Examples of such expressions include: “at least one of A, B, and C”; “one or more of A, B, and C”; “at least one of A, B, or C”; “one or more of A, B, or C”. Unless expressly indicated otherwise, the above expressions encompass any combination of A and/or B and/or C.

[0050] It will be appreciated that some embodiments may be comprised of one or more specialized processors (or “processing devices”) such as microprocessors, digital signal processors, customized processors and field programmable gate arrays (FPGAs) and unique stored program instructions (including both software and firmware) that control the one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of the method and/or apparatus described herein. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of certain of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used.

[0051] Moreover, an embodiment can be implemented as a computer-readable storage medium having computer readable code stored thereon for programming a computer (e.g., comprising a processor) to perform a method as described and claimed herein. Examples of such computer-readable storage mediums include, but are not limited to, a hard disk, a CD-ROM, an optical storage device, a magnetic storage device, a ROM (Read Only Memory), a PROM (Programmable Read Only Memory), an EPROM (Erasable Programmable Read Only Memory), an EEPROM (Electrically Erasable Programmable Read Only Memory) and a Flash memory. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

[0052] The Abstract of the Disclosure is provided to allow the reader to quickly ascertain the nature of the technical disclosure. It is submitted with the understanding that it will not be used to interpret or limit the scope or meaning of the claims. In addition, in the foregoing Detailed Description, it can be seen that various features are grouped together in various embodiments for the purpose of streamlining the disclosure. This method of disclosure is not to be interpreted as reflecting an intention that the claimed embodiments require more features than are expressly recited in each claim. Rather, as the following claims reflect, inventive subject matter lies in less

than all features of a single disclosed embodiment. Thus the following claims are hereby incorporated into the Detailed Description, with each claim standing on its own as a separately claimed subject matter.

Claims

1. An article, comprising: a first portion having a first graphic; a second portion having a second graphic, the second portion being translucent; and a body extending between the first portion and the second portion, the body being deformable to affix the second portion at a position overlaid with the first portion, wherein the second graphic is configured to overlay with the first graphic to define a composite graphic corresponding to the position.
 2. The article of claim 1, wherein the composite graphic is imageable to generate a unique identifier corresponding to the article.
 3. The article of claim 1, wherein the article is a wristband having the first portion at a first end of the body, and the second portion at an opposing end of the body.
 4. The article of claim 3, further comprising: a translucent adhesive disposed on one of the first portion and the second portion, to affix the second portion to the first portion.
 5. The article of claim 1, wherein the first graphic has a first surface area; the second graphic has a second surface area smaller than the first surface area; and the composite graphic is defined by a location of the second graphic within the first surface area.
 6. The article of claim 1, wherein: the second graphic includes a boundary; the first graphic includes a plurality of marks; and the composite graphic is defined by a subset of the plurality of marks of the first graphic encompassed by the boundary.
 7. The article of claim 6, wherein the second graphic includes a second plurality of marks within the boundary; and the composite graphic is further defined by intersections between the marks of the second graphic and the subset of the plurality of marks of the first graphic.
 8. The article of claim 1, wherein the first graphic has a first color, and the second graphic has a second color distinct from the first color.
 9. The article of claim 1, further comprising: a data carrier supported by the body.
 10. A method, comprising: capturing an image of an article having a first graphic overlaid with a second graphic to define a composite graphic, the composite graphic including a boundary of the second graphic, marks of the second graphic, and a portion of a plurality of marks of the first graphic; detecting, from the image, the boundary of the second graphic; generating an identifier based on the composite graphic; and executing an authentication action based on the identifier.
 11. The method of claim 10, wherein executing the authentication action includes: enrolling the identifier; or retrieving a reference identifier, and comparing the reference identifier with the identifier.
 12. The method of claim 10, wherein generating the identifier includes: detecting a background within the boundary of the second graphic; and generating a color value corresponding to the background.
 13. The method of claim 12, further comprising: calibrating the image based on a first predetermined color corresponding to the first graphic, and a second predetermined color corresponding to the second graphic; and generating the color value in response to calibrating the image.
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